

Humidification

Pre-Lab Plan
Unit Operations Lab # **1**
01/23/2020

Commented [DL1]: 29.5/35

Group #2, Thursday, AM section
Diana Awimrin
Leena Daniel
Jay Rawal
Esteban Vazquez

1. Experimental Objective

The objective of this lab is to obtain knowledge of both mass and heat in a wetted wall column by obtaining experimental values for the heat and mass transfer coefficients.

2. Experimental

2.1 Apparatus

Commented [DL2]: 3.5/5

Commented [DL3]: The objective of this experiment is to study the transport of both mass and heat in a wetted wall column and obtain experimental values for the heat and mass transfer coefficients.

Commented [AZ4]: Provide a bit more detail on the conditions and the experiment; heated/unheated air, varying air flowrates, etc.

Commented [DL5]: These values will be obtained through experimental methods and will be compared to values calculated using the theoretical equations.

The system variables that will serve as the independent variables for this experiment include liquid flowrate, air flowrate, and temperature of the incoming air.

Commented [DL6]: 4/5

A brief introduction of the apparatus is needed. You may include the flowsheet of the process.

Commented [AZ7]: Provide text before the figures and tables to introduce them.

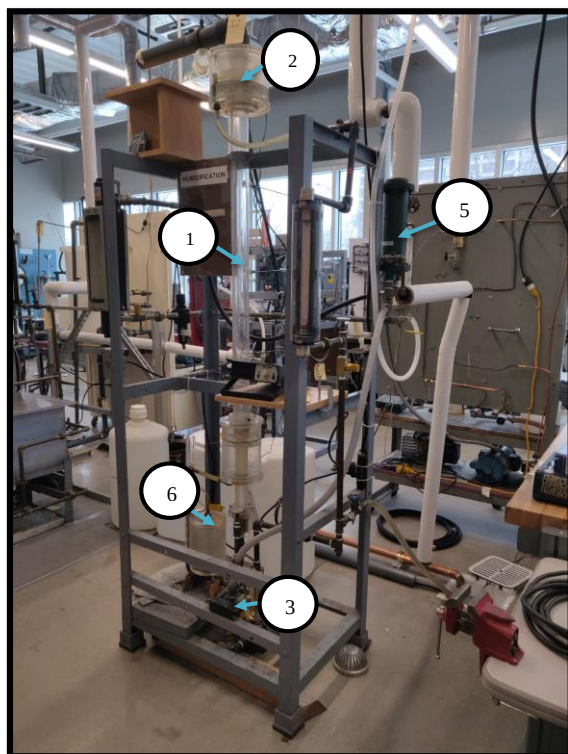


Figure 1: Left Side View of the Apparatus

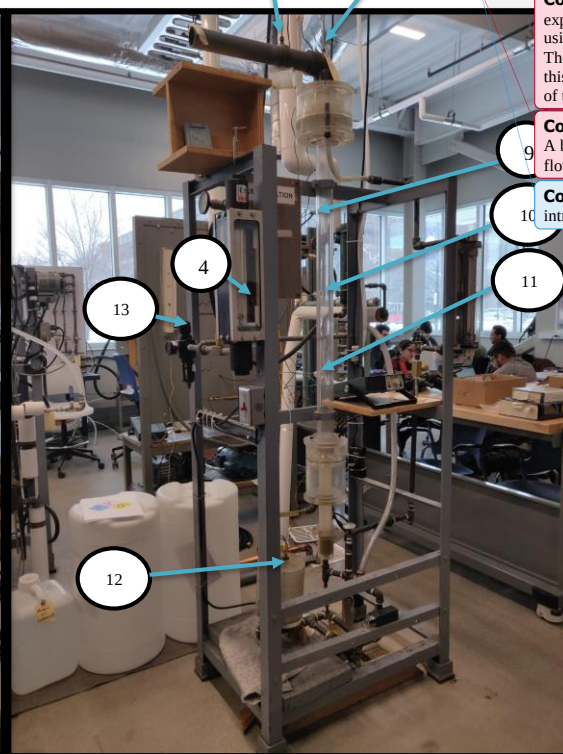


Figure 2: Right Side View of the Apparatus

Commented [DL8]: Caption.
Include other rotameters and the thermocouple reader.

Number	Equipment	Description
1	Wetted wall column humidification unit	Plexi glass tube that is connected to a top and bottom weir with an inlet water and air flow system
2	Holding Tank	Where the water pumped into the wetted-wall column is stored
3	Small Centrifugal Pump	Pumps water into the wetted-wall column
4	Rotameter	Determines the flowrate of the air and has a valve to control flow in system
5	Heat Exchanger	Heats the air coming in and has a valve to control the flow of heated water
6	Reservoir tank	Holds water and has valve to release the water
7	T1 coupling (Wet bulb)	They provide the temperature of water and air through an interface
8	T2 coupling (Dry bulb)	They provide the temperature of water and air through an interface
9	T3 coupling (Top column)	They provide the temperature of water and air through an interface
10	T4 coupling (middle column)	They provide the temperature of water and air through an interface
11	T5 coupling (bottom column)	They provide the temperature of water and air through an interface
12	T6 coupling (reservoir column)	They provide the temperature of water and air through an interface
13	Regulator	Controls air flow into the rotameter

Commented [AZ9]: Also called a weir

Commented [DL10]: Unit and the maximum capacity.

Commented [DL11]: Good.

2.2 Materials and Supplies

Use a table to describe your materials and supplies. Remember that materials and supplies are different than your equipment.

Commented [DL12]: 4.5/5

Materials/Supplies	Brief Description
Water	Water will be used to flow upwards, countercurrent flow with air
Air	Air will be used to flow downwards, countercurrent flow with water
Electronic Hygrometer	Measures the temperature of the air leaving the wetted-wall column

Commented [DL13]: Include the stepladder and the steam.

2.3 Experimental Plan

1. Measure the humidity of the air coming into the column using the wet or dry bulb thermometers or the electronic hygrometer
2. Measure the air temperature to determine the humidity of the air coming in
3. Begin to wet the inside of tube's wall by allowing water to flow at an appropriate rate to allow the wall of the tube to be completely wet
4. Allow unheated air to flow into the column to establish a steady state for the circulation of the water's temperature
5. Measure the steady state temperature values of the water and air at different points in the column
6. Measure the values of the air humidity using the wet and dry thermometers or the electronic hygrometer
7. Repeat steps 1-6 for two other air flowrates
8. Repeat steps 1-6, using the same three air flowrates but using heated air for step 4.
 - a. The air can be heated by using a heat exchanger which uses steam on the shell side to heat the air in two different temperatures.

Commented [DL14]: 8.5/10

Commented [DL15]: How much?

Commented [DL16]: What's the criteria of the steady state?

Commented [DL17]: Be specific.

Commented [DL18]: Be specific.

Project Safety Evaluation

Project Title: Humidification

Date: 01/23/2020

	Yes/No	
Potential electrical hazards?	Yes	If yes, identify conditions: The switch that turns on the pump
Potential mechanical hazards?	No	If yes, identify conditions:
Potential pressure hazards?	No	If yes, identify conditions:
Potential temperature hazards?	Yes	If yes, identify conditions: Steam. High temperatures involving pipes
Hazardous/toxic chemicals involved?	No	If yes, hazards involved:
Gloves required?	Yes	If yes, specify type: Heat resistance gloves
MSDS reviewed by all team members?	Yes	List MSDS sheets reviewed: Air (P-4560-J) and Water
<i>Process Parameters</i>	<i>Max Expected</i>	<i>List location and possible hazards and precautions</i>
Temperature (°C)	90 C	Max is 110C but we will not meet this temperature.
Pressure (psia)	12psia	Max is 12psia because more would cause us to run out. A regulator is attached to the system to allow complete control of air to prevent high pressure build up.
<i>Safety Controls</i>	<i>yes/no</i>	<i>If yes, Provide description</i>
Control Valves	Yes	1. Controlling flow rate and temperature
Surge Protector		2. 15-20 amp surge protector
Regulator		3. A regulator is attached to the system to allow complete control of air to prevent high pressure build up.

I have read relevant background material for the Unit Operations Laboratory entitled: “_____ Humidification_____” and understand the hazards associated with conducting this experiment. I have planned out my experimental work in accordance to standards and acceptable safety practices and will conduct all of my experimental work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.

<u>Diana Awimrin</u>	<u>Jan 23, 2020</u>
signature of team member	date
<u>Leena Daniel</u>	<u>Jan 23, 2020</u>
signature of team member	date
<u>Jay Rawal</u>	<u>Jan 23, 2020</u>
signature of team member	date
<u>Esteban Vazquez</u>	<u>Jan 23, 2020</u>
signature of team member	date

Commented [DL19]: 9/10

Commented [DL20]: Fall from the ladder, or failure of the pump.

Commented [DL21]: House supply air (120 psig max, 40 psig regulated).

Commented [DL22]: CAS number.

Commented [DL23]: 40 psig. House supply air (40 psig).

Commented [AZ24]: Mention that this is for the steam.

Commented [AZ25]: Air system

Commented [AZ26]: Explain for what fluids

Commented [AZ27]: Pump switch also